MS84 Native iDempiere reconciliation and historical replay

Delivered by PR #215, merged 26 September 2026 as `2b5a576`.

Status: corrected native replay completed with zero unexplained data differences at the admitted checkpoints. Broader qualification remains unclaimed.

Customer benefit

Turn the differences found in MS83 into an explainable upgrade path. A reviewer can see where the databases started, which scripts ran, which repairs were necessary, and why each remaining raw difference was accepted.

In beginner terms, both databases begin with comparable data, receive their respective migration instructions, and are checked again before the next release proceeds. A difference must either satisfy a specific reviewed rule or be fixed and measured again. An unexplained difference stops advancement.

Delivered behavior

Direct native calls established that registration works on both engines. The probe restored its affected tables and verified their prior hashes, including the effects of Oracle's independently committed registration transaction.

For the fresh release-13 rerun, one Oracle snapshot supplied 185,275 rows across 911 common base tables. PostgreSQL's 3,807 foreign keys were checked after the load, with zero violations. Canonical source line endings removed differences caused by Windows newlines inside SQL text. Native recompilation resolved the 14 invalid Oracle objects; later checks also validated 2,914 outstanding Oracle foreign keys without changing row data. All 3,826 final release-13 Oracle foreign keys were enabled and validated.

Historical replay began from the official release-3.1 seed, with 843 common tables copied from one source and 3,402 foreign keys checked. It incorporated 51 older prerequisite pairs before the existing inventory and carried admitted release states forward through release 13. Exact Oracle statement dispatch preserves literal text and separates native execution from the static parser.

Measurement	Fresh release-13 experiment	Historical chain
Migration pairs	20 regular + 4 maintenance	1,125 regular
Native script executions / attempts	48	2,255, including 5 recovery attempts
Admitted checkpoints	20 regular + 1 maintenance	20 release checkpoints
Unexplained final data differences	**0**	**0**
Retained allowed final field differences	640	5,540

The historical chain covers all **1,074 original regular inventory pairs** plus the **51 prerequisites**. Its 20 regular pairs overlapping the fresh experiment are not additional unique coverage. Four maintenance pairs remain a separately reported release-13 experiment. The 2,255 attempts include recorded recoveries; earlier failed bootstrap runs are retained separately.

At the final historical checkpoint, 913 common tables were compared: 883 were byte-exact and the remaining raw field differences satisfied explicit rules. Oracle had no invalid objects, compilation errors, unusable indexes or index partitions, disabled constraints or unvalidated constraints. PostgreSQL had no invalid indexes or unvalidated constraints under the retained checks. All 2,914 outstanding historical

Oracle foreign keys were natively validated and complete table row multisets were unchanged. Final foreign-key counts still differ: 3,794 in Oracle and 3,822 in PostgreSQL.

How expected differences are validated

Rules apply only to named fields and observed conditions. They cover fresh, valid, unique UUIDs; clocks within recorded execution windows; registration paths matching the actual executed source; timezone values representing the same instant; and exactly equal finite decimals with different formatting. They do not globally ignore timestamps, trim business text, or use numeric tolerances.

For example, a hypothetical new metadata row may receive different UUIDs on the two engines. That can be admitted only when both IDs satisfy the fresh-row rule. An unexplained change to an existing row's UUID or business value still fails. Both raw values remain available after acceptance.

Earlier accepted differences carry forward only when their witnesses are bound to the actual starting snapshots and both raw values remain unchanged. Unknown tables, columns, row keys or differences block admission. Original rejected checkpoints remain beside their reviewed successors.

Repairs and claim boundaries

The replay required explicit corrections, including old-seed prerequisites, timestamp precision and literal handling, blank-value predicates, missing dictionary rows, and constraint/index operations incompatible with the observed starting schema. Native probes, original failures and repair evidence are retained. The nine earlier application-table findings are resolved on this corrected path.

This proves agreement of complete common base-table data at the declared checkpoints after documented repairs. Historical release checkpoints do not provide isolated equivalence proofs for each of the 1,125 script pairs, and the result is not unmodified-upstream-script equivalence.

Schema equivalence, full application equivalence, platform qualification and Oracle behavior-catalog native conformance remain unclaimed. The differing foreign-key counts are visible rather than waived. Static calibration remains **780 / 111,293 decided SQL units**; native replay does not increase that count. Running an old seed on Oracle 26ai also does not qualify Oracle 11g or 19c.

Evidence and validation

The reconciliation report, release results, sealed receipt and evidence archive bind source pins, rules, native observations, repairs and validity results. The archive contains 9,671 members, including 34 research controllers. Complete all-row snapshots remain in ignored local storage; published hashes cannot reconstruct those rows. Integrity hashes are not independent attestation.

The affected suite ran 133 tests with two local skips; the optional pinned-source replay was run separately and passed. Publication integrity, controller syntax and credential scans passed. A Windows-path portability issue in the publication verifier was corrected before merge. All **21 PR checks passed**, including Windows and Linux calibration, pinned-source replay and broader regressions.

Execution used local Oracle 26ai Free and PostgreSQL 15. No GCP resources were started, and the task's database containers were stopped with data retained.

What this enables next

The retained checkpoints and documented repairs provide a concrete baseline for repeatable upgrade tooling and customer-specific assessment. Application journeys, structural/schema differences and operational platform dimensions need their own evidence before this result can support a broader production-readiness claim.